Class 9 Mathematics — Exercise 4.7

Basic Concept

Square Root of an Expression

- The square root of 16 is ± 4, because $(+4)^2 = 16$ and $(-4)^2 = 16$. So the square root of an expression is always written with $\pm$.
- **By factorization:** write the expression as a complete square $(\quad)^2$ using $(a \pm b)^2 = a^2 \pm 2ab + b^2$, then take the square root on both sides.
- **By division method:**
  1. Arrange the expression in descending (or ascending) powers.
  2. Take the square root of the first term; this is the first term of the root. Subtract its square.
  3. **Double** the root found so far; divide the first term of the remainder by the first term of this double to get the next term of the root. Add it to the divisor, multiply, and subtract.
  4. Repeat until the remainder is 0.

Question 1

Find the square root by factorization: $16y^2 - 56y + 49$

($\because (a-b)^2 = a^2 - 2ab + b^2, \quad \sqrt{16} = 4$)

$$16y^2 - 56y + 49 = (4y)^2 - 2(4y)(7) + (7)^2 = (4y-7)^2$$

Taking square root on both sides:

$$\sqrt{16y^2 - 56y + 49} = \sqrt{(4y-7)^2} = \boxed{\pm(4y-7)} \text{ Ans}$$

Question 2

$$25a^4 - 30a^3 + 9a^2$$

1st Method:

$$25a^4 - 30a^3 + 9a^2 = a^2(25a^2 - 30a + 9) = a^2[(5a)^2 - 2(5a)(3) + (3)^2] = a^2(5a-3)^2$$

Taking square root on both sides:

$$\sqrt{25a^4 - 30a^3 + 9a^2} = \sqrt{a^2(5a-3)^2} = \boxed{\pm a(5a-3)} \text{ Ans}$$

2nd Method:

$$25a^4 - 30a^3 + 9a^2 = (5a^2)^2 - 2(5a^2)(3a) + (3a)^2 = (5a^2 - 3a)^2$$

$$\sqrt{25a^4 - 30a^3 + 9a^2} = \pm(5a^2 - 3a) = \pm a(5a - 3)$$

Question 3

$$\left(x^2 - \frac{1}{x^2}\right)^2 + 4\left(x^2 - \frac{1}{x^2}\right) + 4, \qquad x \neq 0$$

$(\because\ a^2 + 2ab + b^2 = (a+b)^2)$

$$= \left(x^2 - \frac{1}{x^2}\right)^2 + 2\left(x^2 - \frac{1}{x^2}\right)(2) + (2)^2 = \left(x^2 - \frac{1}{x^2} + 2\right)^2$$

Taking square root on both sides:

$$\sqrt{\left(x^2 - \frac{1}{x^2}\right)^2 + 4\left(x^2 - \frac{1}{x^2}\right) + 4} = \boxed{\pm\left(x^2 - \frac{1}{x^2} + 2\right)} \text{ Ans}$$

Question 4

$$\left(a^2 + \frac{1}{a^2}\right) - 8\left(a - \frac{1}{a}\right) + 14, \qquad a \neq 0$$

$$= a^2 + \frac{1}{a^2} - 8\left(a - \frac{1}{a}\right) + 14 + 2 - 2$$

$$= \left(a^2 + \frac{1}{a^2} - 2\right) - 8\left(a - \frac{1}{a}\right) + 16$$

$$= \left[(a)^2 + \left(\frac{1}{a}\right)^2 - 2(a)\left(\frac{1}{a}\right)\right] - 8\left(a - \frac{1}{a}\right) + (4)^2$$

$$= \left(a - \frac{1}{a}\right)^2 - 2\left(a - \frac{1}{a}\right)(4) + (4)^2$$

$$= \left(a - \frac{1}{a} - 4\right)^2$$

Taking square root on both sides:

$$\sqrt{\left(a^2 + \frac{1}{a^2}\right) - 8\left(a - \frac{1}{a}\right) + 14} = \boxed{\pm\left(a - \frac{1}{a} - 4\right)} \text{ Ans}$$

Question 5

$$(a+2)(a+4)(a+6)(a+8)+16$$

Pairing: $2+8=10$ and $4+6=10$

$$\begin{aligned}
&= (a+2)(a+8)(a+4)(a+6)+16 \\
&= [a(a+8)+2(a+8)][a(a+6)+4(a+6)]+16 \\
&= (a^2+10a+16)(a^2+10a+24)+16
\end{aligned}$$

Let $a^2+10a=x$

$$\begin{aligned}
&= (x+16)(x+24)+16 \\
&= x(x+24)+16(x+24)+16 \\
&= x^2+24x+16x+384+16 \\
&= x^2+40x+400 \\
&= x^2+20x+20x+400 \\
&= x(x+20)+20(x+20) \\
&= (x+20)^2
\end{aligned}$$

Put value of x:

$$= (a^2+10a+20)^2$$

Taking square root on both sides:

$$\sqrt{(a+2)(a+4)(a+6)(a+8)+16} = \boxed{\pm(a^2+10a+20)} \text{ Ans}$$

Questions 6 to 14

Find the square root of the following by division method.

Question 6

$$x^4+8x^3+20x^2+16x+4$$

- $\sqrt{x^4}=x^2 \Rightarrow$ first term of root $=x^2$. Subtract $(x^2)^2=x^4$.
- Divisor $=2(x^2)+(4x)=2x^2+4x$; $(2x^2+4x)(4x)=8x^3+16x^2$; remainder $=4x^2+16x+4$
- Divisor $=2(x^2+4x)+(2)=2x^2+8x+2$; $(2x^2+8x+2)(2)=4x^2+16x+4$; remainder $=0$

Remainder $=0$

$$\text{Square root} = \boxed{\pm\,(x^2+4x+2)} \text{ Ans}$$

Question 7

$$x^4 + 10x^3 + 31x^2 + 30x + 9$$

- $\sqrt{x^4} = x^2 \Rightarrow$ first term of root $= x^2$. Subtract $(x^2)^2 = x^4$.
- Divisor $= 2(x^2) + (5x) = 2x^2 + 5x$; $(2x^2 + 5x)(5x) = 10x^3 + 25x^2$; remainder $= 6x^2 + 30x + 9$
- Divisor $= 2(x^2 + 5x) + (3) = 2x^2 + 10x + 3$; $(2x^2 + 10x + 3)(3) = 6x^2 + 30x + 9$; remainder $= 0$

Remainder $= 0$

$$\text{Square root} = \boxed{\pm\,(x^2 + 5x + 3)} \text{ Ans}$$

Question 8

$$49b^4 + 18a^2b^2 + 4a^3b + a^4 + 28ab^3$$

- $\sqrt{49b^4} = 7b^2 \Rightarrow$ first term of root $= 7b^2$. Subtract $(7b^2)^2 = 49b^4$.
- Divisor $= 2(7b^2)+(2ab) = 2ab+14b^2$; $(2ab+14b^2)(2ab) = 4a^2b^2+28ab^3$; remainder $= a^4 + 4a^3b + 14a^2b^2$
- Divisor $= 2(2ab+7b^2)+(a^2) = a^2+4ab+14b^2$; $(a^2+4ab+14b^2)(a^2) = a^4+4a^3b+14a^2b^2$; remainder $= 0$

Remainder $= 0$

$$\text{Square root} = \boxed{\pm\,(7b^2 + 2ab + a^2)} \text{ Ans}$$

Question 9

$$4x^4 - 12x^3 + 29x^2 - 30x + 25$$

- $\sqrt{4x^4} = 2x^2 \Rightarrow$ first term of root $= 2x^2$. Subtract $(2x^2)^2 = 4x^4$.
- Divisor $= 2(2x^2) + (-3x) = 4x^2 - 3x$; $(4x^2 - 3x)(-3x) = -12x^3 + 9x^2$; remainder $= 20x^2 - 30x + 25$
- Divisor $= 2(2x^2 - 3x) + (5) = 4x^2 - 6x + 5$; $(4x^2 - 6x + 5)(5) = 20x^2 - 30x + 25$; remainder $= 0$

Remainder $= 0$

$$\text{Square root} = \boxed{\pm\,(2x^2 - 3x + 5)} \text{ Ans}$$

Question 10

$$1 - 10x + 27x^2 - 10x^3 + x^4$$

- $\sqrt{1} = 1 \Rightarrow$ first term of root $= 1$. Subtract $(1)^2 = 1$.
- Divisor $= 2(1) + (-5x) = 2 - 5x$; $(2 - 5x)(-5x) = 25x^2 - 10x$; remainder $= x^4 - 10x^3 + 2x^2$
- Divisor $= 2(1 - 5x) + (x^2) = x^2 - 10x + 2$; $(x^2 - 10x + 2)(x^2) = x^4 - 10x^3 + 2x^2$; remainder $= 0$

Remainder $= 0$

$$\text{Square root} = \boxed{\pm\,(1 - 5x + x^2)} \text{ Ans}$$

Question 11

$$x^4 + \frac{1}{x^4} + 4x^2 - \frac{4}{x^2} + 2, \quad x \neq 0$$

- $\sqrt{x^4} = x^2 \Rightarrow$ first term of root $= x^2$. Subtract $(x^2)^2 = x^4$.
- Divisor $= 2(x^2) + (2) = 2x^2 + 2$; $(2x^2 + 2)(2) = 4x^2 + 4$; remainder $= -2 - \frac{4}{x^2} + \frac{1}{x^4}$
- Divisor $= 2(x^2 + 2) + (-\frac{1}{x^2}) = 2x^2 + 4 - \frac{1}{x^2}$; $(2x^2 + 4 - \frac{1}{x^2})(-\frac{1}{x^2}) = -2 - \frac{4}{x^2} + \frac{1}{x^4}$; remainder $= 0$

Remainder $= 0$

$$\text{Square root} = \boxed{\pm\left(x^2 + 2 - \frac{1}{x^2}\right)} \text{ Ans}$$

Question 12

$$a^2 - 8a + 2 + \frac{56}{a} + \frac{49}{a^2}, \quad a \neq 0$$

- $\sqrt{a^2} = a \Rightarrow$ first term of root $= a$. Subtract $(a)^2 = a^2$.
- Divisor $= 2(a) + (-4) = 2a - 4$; $(2a - 4)(-4) = 16 - 8a$; remainder $= -14 + \frac{56}{a} + \frac{49}{a^2}$
- Divisor $= 2(a - 4) + (-\frac{7}{a}) = 2a - 8 - \frac{7}{a}$; $(2a - 8 - \frac{7}{a})(-\frac{7}{a}) = -14 + \frac{56}{a} + \frac{49}{a^2}$; remainder $= 0$

Remainder $= 0$

$$\text{Square root} = \boxed{\pm\left(a - 4 - \frac{7}{a}\right)} \text{ Ans}$$

Question 13

$$x^4 - 2x^3 + \frac{3x^2}{2} - \frac{x}{2} + \frac{1}{16}$$

- $\sqrt{x^4} = x^2 \Rightarrow$ first term of root $= x^2$. Subtract $(x^2)^2 = x^4$.
- Divisor $= 2(x^2) + (-x) = 2x^2 - x$; $(2x^2 - x)(-x) = -2x^3 + x^2$; remainder $= \frac{x^2}{2} - \frac{x}{2} + \frac{1}{16}$
- Divisor $= 2(x^2 - x) + (\frac{1}{4}) = 2x^2 - 2x + \frac{1}{4}$; $(2x^2 - 2x + \frac{1}{4})(\frac{1}{4}) = \frac{x^2}{2} - \frac{x}{2} + \frac{1}{16}$; remainder $= 0$

Remainder $= 0$

$$\text{Square root} = \boxed{\pm\left(x^2 - x + \frac{1}{4}\right)} \text{ Ans}$$

Question 14

$$4x^4 + 32x^2 + 96 + \frac{64}{x^4} + \frac{128}{x^2}$$

- $\sqrt{4x^4} = 2x^2 \Rightarrow$ first term of root $= 2x^2$. Subtract $(2x^2)^2 = 4x^4$.
- Divisor $= 2(2x^2) + (8) = 4x^2 + 8$; $(4x^2 + 8)(8) = 32x^2 + 64$; remainder $= 32 + \frac{128}{x^2} + \frac{64}{x^4}$
- Divisor $= 2(2x^2 + 8) + (\frac{8}{x^2}) = 4x^2 + 16 + \frac{8}{x^2}$; $(4x^2 + 16 + \frac{8}{x^2})(\frac{8}{x^2}) = 32 + \frac{128}{x^2} + \frac{64}{x^4}$; remainder $= 0$

Remainder $= 0$

$$\text{Square root} = \boxed{\pm\left(2x^2 + 8 + \frac{8}{x^2}\right)} \text{ Ans}$$

Question 15

To make $a^4 - 10a^3 + 27a^2 - 9a + 2$ a perfect square: (i) what should be added to it? (ii) what should be subtracted from it? (iii) what will be the value of a?

Division method:

- $\sqrt{a^4} = a^2$. Subtract a^4: remainder $= -10a^3 + 27a^2 - 9a + 2$
- Divisor $= 2a^2 - 5a$; $(2a^2 - 5a)(-5a) = -10a^3 + 25a^2$; remainder $= 2a^2 - 9a + 2$
- Divisor $= 2a^2 - 10a + 1$; $(2a^2 - 10a + 1)(1) = 2a^2 - 10a + 1$; remainder $= a + 1$

Root $= a^2 - 5a + 1$, Remainder $= a + 1$

(i) To make it a perfect square, we should **add** $\boxed{-(a+1) = -a - 1}$

(ii) We should **subtract** $\boxed{a + 1}$

(iii) For a perfect square the remainder must be zero: $a + 1 = 0 \Rightarrow \boxed{a = -1}$

Question 16

Find the value of p and q if $x^4 - 12x^3 + px + q$ is a complete square.

Write it as $x^4 - 12x^3 + 0x^2 + px + q$.

- $\sqrt{x^4} = x^2$. Subtract x^4: remainder $= -12x^3 + 0x^2 + px + q$
- Divisor $= 2x^2 - 6x$; $(2x^2 - 6x)(-6x) = -12x^3 + 36x^2$; remainder $= -36x^2 + px + q$
- Divisor $= 2x^2 - 12x - 18$; $(2x^2 - 12x - 18)(-18) = -36x^2 + 216x + 324$
- Remainder $= (p - 216)x + (q - 324)$

For a complete square, remainder $= 0$:

$$p - 216 = 0 \Rightarrow \boxed{p = 216} \qquad q - 324 = 0 \Rightarrow \boxed{q = 324}$$

Question 17

For what value of k does the expression $y^4 + 4y^2 + k + \frac{8}{y^2} + \frac{4}{y^4}$ become a perfect square, where $y \neq 0$?

- $\sqrt{y^4} = y^2$. Subtract y^4: remainder $= 4y^2 + k + \frac{8}{y^2} + \frac{4}{y^4}$
- Divisor $= 2y^2 + 2$; $(2y^2 + 2)(2) = 4y^2 + 4$; remainder $= (k - 4) + \frac{8}{y^2} + \frac{4}{y^4}$
- Divisor $= 2y^2 + 4 + \frac{2}{y^2}$; $\left(2y^2 + 4 + \frac{2}{y^2}\right)\left(\frac{2}{y^2}\right) = 4 + \frac{8}{y^2} + \frac{4}{y^4}$
- Remainder $= k - 4 - 4 = k - 8$

For a perfect square, remainder $= 0$:

$$k - 8 = 0 \Rightarrow \boxed{k = 8} \text{ Ans}$$